

Blockchain Experiment 9

AIM: Implement the Private Ethereum Blockchain using Geth

THEORY:

Q1: What is Geth?

Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go programming language. It allows users to run an Ethereum node, mine Ether, and interact with the blockchain through JSON-RPC, command-line, or JavaScript consoles.

Q2: Significance of a Private Ethereum Network

Setting up a Private Ethereum Network involves creating a customized blockchain environment isolated from the public Ethereum network. It enables developers to test smart contracts, consensus mechanisms, and network behavior without incurring gas costs or depending on public validators.

Q3: Steps for creating a Private Ethereum Network

Steps Involved:

- **Choosing a Network ID:**
Each private blockchain must have a unique network ID to differentiate it from public or other private networks.
- **Choosing a Consensus Algorithm:**
Ethereum supports different consensus mechanisms such as Proof of Work (PoW) and Proof of Authority (PoA). In private networks, Clique (PoA) is often used for faster block confirmation.
- **Creating a Genesis Block:**
The genesis block is the first block of the blockchain, defining network parameters such as difficulty, gas limit, initial account balances, and consensus settings.
- **Initializing the Geth Database:**
The Geth init command initializes the blockchain using the genesis.json file, preparing the data directory for node operation.
- **Setting up Networking:**
Nodes communicate using enodes (Ethereum Node URLs) that contain IP addresses and public keys for peer discovery and connection.
- **Running the Member Nodes:**
Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.
- **Running a Signer (in Clique):**
In the Clique consensus, designated signers validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

NOTE: Download the genesis file and edit the account details, including the public keys of all participating peers in the network.

PROCEDURE

Step 1: Install Geth

```
wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
tar -xvf geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
mkdir -p ~/bin
cp geth-linux-amd64-1.13.14-2bd6bd01/geth ~/bin/geth113
export PATH="$HOME/bin:$PATH"
geth113 version
```

```
Get:5 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main amd64 ripdump amd64 1.17.2+build8480+noble [904 kB]
Get:6 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main amd64 ethereum amd64 1.17.2+build8480+noble [1,442 B]
Fetched 51.1 MB in 2min 2s (417 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package abigen.
(Reading database ... 125206 files and directories currently installed.)
Preparing to unpack .../0-abigen_1.17.2+build8480+noble_amd64.deb ...
Unpacking abigen (1.17.2+build8480+noble) ...
Selecting previously unselected package clef.
Preparing to unpack .../1-clef_1.17.2+build8480+noble_amd64.deb ...
Unpacking clef (1.17.2+build8480+noble) ...
Selecting previously unselected package evm.
Preparing to unpack .../2-evm_1.17.2+build8480+noble_amd64.deb ...
Unpacking evm (1.17.2+build8480+noble) ...
Selecting previously unselected package geth.
Preparing to unpack .../3-geth_1.17.2+build8480+noble_amd64.deb ...
Unpacking geth (1.17.2+build8480+noble) ...
Selecting previously unselected package ripdump.
Preparing to unpack .../4-ripdump_1.17.2+build8480+noble_amd64.deb ...
Unpacking ripdump (1.17.2+build8480+noble) ...
Selecting previously unselected package ethereum.
Preparing to unpack .../5-ethereum_1.17.2+build8480+noble_amd64.deb ...
Unpacking ethereum (1.17.2+build8480+noble) ...
Setting up ripdump (1.17.2+build8480+noble) ...
Setting up clef (1.17.2+build8480+noble) ...
Setting up geth (1.17.2+build8480+noble) ...
Setting up evm (1.17.2+build8480+noble) ...
Setting up abigen (1.17.2+build8480+noble) ...
Setting up ethereum (1.17.2+build8480+noble) ...
Geth
Version: 1.17.2-stable
Git Commit: be4dc0c4be2fe316dbdd0a73e48421f64978232f
Architecture: amd64
Go Version: go1.25.7
Operating System: linux
GOPATH=/home/g2022_sneha_patra/gopath:/google/gopath
GOROOT=
g2022_sneha_patra@cloudshell:~$
```

Step 2: Create folders and accounts for both nodes

```
GOROOT=/home/g2022_sneha_patra/gopath:/google/gopath
GOROOT=
g2022_sneha_patra@cloudshell:~$ mkdir -p ~/eth-private/node1 ~/eth-private/node2
geth --datadir ~/eth-private/node1 account new
geth --datadir ~/eth-private/node2 account new
INFO [04-09T14:14:13.136] Maximal peer count: 2560
INFO [04-09T14:14:13.140] Smartcard socket not found, disabling: ""
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key: 0x728383386d0a032e1aE23010d6675D0ee6d37547
Path of the secret key file: /home/g2022_sneha_patra/eth-private/node1/keystore/UTC--2026-04-09T14-14-30.606120975S--728383386d0a032e1aE23010d6675D0ee6d37547

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!

INFO [04-09T14:14:35.401] Maximal peer count: 2560
INFO [04-09T14:14:35.405] Smartcard socket not found, disabling: ""
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key: 0x63c60A32954Ae330527d140657285282F34080CF
Path of the secret key file: /home/g2022_sneha_patra/eth-private/node2/keystore/UTC--2026-04-09T14-14-32.386529138Z--63c60A32954Ae330527d140657285282F34080CF

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!

g2022_sneha_patra@cloudshell:~$
```

```
g2022_sneha_patra@cloudshell:~$ echo "Sneha@100" > ~/eth-private/node1/password.txt
echo "Sneha@100" > ~/eth-private/node2/password.txt
g2022_sneha_patra@cloudshell:~$
```

```
NODE1="PASTE_NODE1_ADDRESS_WITHOUT_0x_HERE"
NODE2="PASTE_NODE2_ADDRESS_WITHOUT_0x_HERE"
```

[illegible]

Step 6: Run Node 1 (signer)

```

g0222_sasha@cloudshell:~$ geth --details ~/eth-private/node1 \
--networkid 1234 \
--port 30303 \
--http --http.port 8545 \
--unlock 0x0[MODE1] \
--password ~/eth-private/node1/password.txt \
--mine \
--miner.etherbase 0x0[MODE1] \
--allow-insecure-unlock \
console
[2019-04-05T13:09:33.536] Maximum peer count: 30
[2019-04-05T13:09:33.550] Smartcard socket not found, disabling
[2019-04-05T13:09:33.563] Set global gas cap
[2019-04-05T13:09:33.564] Initializing the xss library
[2019-04-05T13:09:33.703] Allocated trie memory caches
[2019-04-05T13:09:33.704] Using petname on the backing database
[2019-04-05T13:09:33.704] Allocated cache and file handles
[2019-04-05T13:09:33.743] Opened ancient database
[2019-04-05T13:09:33.743] State scheme set to already existing
[2019-04-05T13:09:33.744] Initializing Ethereum protocol
[2019-04-05T13:09:33.745]
-----
[2019-04-05T13:09:33.745] Chain ID: 1234 (unknown)
[2019-04-05T13:09:33.745] Consensus: Clique (proof-of-authority)
[2019-04-05T13:09:33.745]
[2019-04-05T13:09:33.745] Free-merge hard forks: Götter basist1
[2019-04-05T13:09:33.745] - Byzantium
[2019-04-05T13:09:33.745] - Tangerine Whistle (EIP 150)
[2019-04-05T13:09:33.745] - Spurious Dragon1 (EIP 150)
[2019-04-05T13:09:33.745] - Spurious Dragon2 (EIP 158)
[2019-04-05T13:09:33.745] - Byzantium
[2019-04-05T13:09:33.745] - Constantinople
[2019-04-05T13:09:33.745] - Petersburg
[2019-04-05T13:09:33.745] - Istanbul

```

Step 7: Open new terminal tab and run Node 2 (member)

[illegible]

```
[INFO] [04-05:13:16:40.850] Websocket enabled
[INFO] [04-05:13:16:40.850] HTTP server started
[INFO] [04-05:13:16:42.636] Unlocked account
WARN [04-05:13:16:42.698] Served eth_coinbase
Welcome to the Geth Javascript console!

instance: Geth/v1.11.14-stable-3dd8d80f/linux-amd64/go1.21.6
at block: 0 (Thu Jan 01 1970 00:00:00 GMT+0000 (UTC))
datadir: /home/g0blin/.geth/geth-eth-private/node2
modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 taproot:1.0 web3:1.0

To exit, press ctrl-d or type exit
> [INFO] [04-05:13:16:43.591] new local node record
[INFO] [04-05:13:16:50.991] Looking for peers
[INFO] [04-05:13:17:01.133] Looking for peers
[INFO] [04-05:13:17:11.176] Looking for peers
[INFO] [04-05:13:17:21.301] Looking for peers
[INFO] [04-05:13:17:31.368] Looking for peers
[INFO] [04-05:13:17:41.668] Looking for peers
[INFO] [04-05:13:17:51.711] Looking for peers
[INFO] [04-05:13:18:01.754] Looking for peers
[INFO] [04-05:13:18:11.797] Looking for peers
[INFO] [04-05:13:18:21.840] Looking for peers
[INFO] [04-05:13:18:31.883] Looking for peers
[INFO] [04-05:13:18:41.926] Looking for peers
[INFO] [04-05:13:18:51.969] Looking for peers
[INFO] [04-05:13:19:02.012] Looking for peers
[INFO] [04-05:13:19:12.055] Looking for peers
[INFO] [04-05:13:19:22.098] Looking for peers
[INFO] [04-05:13:19:32.141] Looking for peers
[INFO] [04-05:13:19:42.184] Looking for peers
[INFO] [04-05:13:19:52.227] Looking for peers
[INFO] [04-05:13:20:02.270] Looking for peers
[INFO] [04-05:13:20:12.313] Looking for peers
[INFO] [04-05:13:20:22.356] Looking for peers
[INFO] [04-05:13:20:32.399] Looking for peers
[INFO] [04-05:13:20:42.442] Looking for peers
[INFO] [04-05:13:20:52.485] Looking for peers
[INFO] [04-05:13:21:02.528] Looking for peers
[INFO] [04-05:13:21:12.571] Looking for peers
[INFO] [04-05:13:21:22.614] Looking for peers
[INFO] [04-05:13:21:32.657] Looking for peers
[INFO] [04-05:13:21:42.700] Looking for peers
[INFO] [04-05:13:21:52.743] Looking for peers
[INFO] [04-05:13:22:02.786] Looking for peers
[INFO] [04-05:13:22:12.829] Looking for peers
[INFO] [04-05:13:22:22.872] Looking for peers
[INFO] [04-05:13:22:32.915] Looking for peers
[INFO] [04-05:13:22:42.958] Looking for peers
[INFO] [04-05:13:22:52.001] Looking for peers
[INFO] [04-05:13:23:02.044] Looking for peers
[INFO] [04-05:13:23:12.087] Looking for peers
[INFO] [04-05:13:23:22.130] Looking for peers
[INFO] [04-05:13:23:32.173] Looking for peers
[INFO] [04-05:13:23:42.216] Looking for peers
[INFO] [04-05:13:23:52.259] Looking for peers
[INFO] [04-05:13:24:02.302] Looking for peers
[INFO] [04-05:13:24:12.345] Looking for peers
[INFO] [04-05:13:24:22.388] Looking for peers
[INFO] [04-05:13:24:32.431] Looking for peers
[INFO] [04-05:13:24:42.474] Looking for peers
[INFO] [04-05:13:24:52.517] Looking for peers
[INFO] [04-05:13:25:02.560] Looking for peers
[INFO] [04-05:13:25:12.603] Looking for peers
[INFO] [04-05:13:25:22.646] Looking for peers
[INFO] [04-05:13:25:32.689] Looking for peers
[INFO] [04-05:13:25:42.732] Looking for peers
[INFO] [04-05:13:25:52.775] Looking for peers
[INFO] [04-05:13:26:02.818] Looking for peers
[INFO] [04-05:13:26:12.861] Looking for peers
[INFO] [04-05:13:26:22.904] Looking for peers
[INFO] [04-05:13:26:32.947] Looking for peers
[INFO] [04-05:13:26:42.990] Looking for peers
[INFO] [04-05:13:26:53.033] Looking for peers
[INFO] [04-05:13:27:03.076] Looking for peers
[INFO] [04-05:13:27:13.119] Looking for peers
[INFO] [04-05:13:27:23.162] Looking for peers
[INFO] [04-05:13:27:33.205] Looking for peers
[INFO] [04-05:13:27:43.248] Looking for peers
[INFO] [04-05:13:27:53.291] Looking for peers
[INFO] [04-05:13:28:03.334] Looking for peers
[INFO] [04-05:13:28:13.377] Looking for peers
[INFO] [04-05:13:28:23.420] Looking for peers
[INFO] [04-05:13:28:33.463] Looking for peers
[INFO] [04-05:13:28:43.506] Looking for peers
[INFO] [04-05:13:28:53.549] Looking for peers
[INFO] [04-05:13:29:03.592] Looking for peers
[INFO] [04-05:13:29:13.635] Looking for peers
[INFO] [04-05:13:29:23.678] Looking for peers
[INFO] [04-05:13:29:33.721] Looking for peers
[INFO] [04-05:13:29:43.764] Looking for peers
[INFO] [04-05:13:29:53.807] Looking for peers
[INFO] [04-05:13:30:03.850] Looking for peers
[INFO] [04-05:13:30:13.893] Looking for peers
[INFO] [04-05:13:30:23.936] Looking for peers
[INFO] [04-05:13:30:33.979] Looking for peers
[INFO] [04-05:13:30:44.022] Looking for peers
[INFO] [04-05:13:30:54.065] Looking for peers
[INFO] [04-05:13:31:04.108] Looking for peers
[INFO] [04-05:13:31:14.151] Looking for peers
[INFO] [04-05:13:31:24.194] Looking for peers
[INFO] [04-05:13:31:34.237] Looking for peers
[INFO] [04-05:13:31:44.280] Looking for peers
[INFO] [04-05:13:31:54.323] Looking for peers
[INFO] [04-05:13:32:04.366] Looking for peers
[INFO] [04-05:13:32:14.409] Looking for peers
[INFO] [04-05:13:32:24.452] Looking for peers
[INFO] [04-05:13:32:34.495] Looking for peers
[INFO] [04-05:13:32:44.538] Looking for peers
[INFO] [04-05:13:32:54.581] Looking for peers
[INFO] [04-05:13:33:04.624] Looking for peers
[INFO] [04-05:13:33:14.667] Looking for peers
[INFO] [04-05:13:33:24.710] Looking for peers
[INFO] [04-05:13:33:34.753] Looking for peers
[INFO] [04-05:13:33:44.796] Looking for peers
[INFO] [04-05:13:33:54.839] Looking for peers
[INFO] [04-05:13:34:04.882] Looking for peers
[INFO] [04-05:13:34:14.925] Looking for peers
[INFO] [04-05:13:34:24.968] Looking for peers
[INFO] [04-05:13:34:35.011] Looking for peers
[INFO] [04-05:13:34:45.054] Looking for peers
[INFO] [04-05:13:34:55.097] Looking for peers
[INFO] [04-05:13:35:05.140] Looking for peers
[INFO] [04-05:13:35:15.183] Looking for peers
[INFO] [04-05:13:35:25.226] Looking for peers
[INFO] [04-05:13:35:35.269] Looking for peers
[INFO] [04-05:13:35:45.312] Looking for peers
[INFO] [04-05:13:35:55.355] Looking for peers
[INFO] [04-05:13:36:05.398] Looking for peers
[INFO] [04-05:13:36:15.441] Looking for peers
[INFO] [04-05:13:36:25.484] Looking for peers
[INFO] [04-05:13:36:35.527] Looking for peers
[INFO] [04-05:13:36:45.570] Looking for peers
[INFO] [04-05:13:36:55.613] Looking for peers
[INFO] [04-05:13:37:05.656] Looking for peers
[INFO] [04-05:13:37:15.699] Looking for peers
[INFO] [04-05:13:37:25.742] Looking for peers
[INFO] [04-05:13:37:35.785] Looking for peers
[INFO] [04-05:13:37:45.828] Looking for peers
[INFO] [04-05:13:37:55.871] Looking for peers
[INFO] [04-05:13:38:05.914] Looking for peers
[INFO] [04-05:13:38:15.957] Looking for peers
[INFO] [04-05:13:38:25.000] Looking for peers
[INFO] [04-05:13:38:35.043] Looking for peers
[INFO] [04-05:13:38:45.086] Looking for peers
[INFO] [04-05:13:38:55.129] Looking for peers
[INFO] [04-05:13:39:05.172] Looking for peers
[INFO] [04-05:13:39:15.215] Looking for peers
[INFO] [04-05:13:39:25.258] Looking for peers
[INFO] [04-05:13:39:35.301] Looking for peers
[INFO] [04-05:13:39:45.344] Looking for peers
[INFO] [04-05:13:39:55.387] Looking for peers
[INFO] [04-05:13:40:05.430] Looking for peers
[INFO] [04-05:13:
```

Step 8: Connect the two nodes

In Node 1 console, get its enode

```
> admin.modelInfo.encode
```

Now, switch to Node 2 console and paste the Node 1 enode

```

C:\msdoin\ad3penc>cmd.exe //741f40155517401562c024f45432fa21961c16959603264707008a00b13f0162093d16e99d334530308671d70992177c00d2xLw9tcc90453a3a721e0c0999608,143,130,399,30103"

```

In Node 2 console, check they are connected

[illegible]

Step 9: Send a transaction

In Node 1 console, send transaction to Node 2

[illegible]

and check the balance of Node 2

```
> eth.getBalance("0x5fd63344c1f40712A9A85d9d7e2B304a6C558716")
1010000000000000000000
```

CONCLUSION:

Through this experiment, two smart contracts - OrderRecord and ReviewVerification, were successfully created and deployed on the Ethereum Sepolia test network using Solidity and Remix IDE. The experiment demonstrated how blockchain technology can be integrated into a full-stack web application to permanently and transparently record order transactions and product reviews. The use of the Sepolia test network and MetaMask allowed safe testing without real funds. Overall, this experiment provided practical understanding of designing